

Student Mental Health Indicators

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Statistical Methods & Applications I

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1 Introduction & Background

1.1 Background & Research Goal

Mental health is an important area of research, especially among student populations. This paper focuses on the factors that may contribute to depression in students. Specifically, we want to understand what kind of students are more prone to depression by analyzing different risk factors and demographic information. This research is relevant and necessary because depression is a serious mental health condition that affects the daily life of many people. As students ourselves, we are interested in any clear indicators that a student is more likely to have depression. Ultimately, our goal is to be able not only to understand and analyze depression levels but also to predict them. This paper will focus on the former and lay groundwork for future predictive analysis. This can inform strategies for the implementation of mental health resources in educational communities to improve the lives of students.

1.2 Data Source

Our dataset, “Student Depression Dataset” by Adil Shamim covers a large range of information concerning depression in students from India. The data is in a .csv format with different features such as demographics, academic indicators, lifestyle, etc. The target variable in this paper is “Depression_Status”, which is a binary variable (Yes = 1, No = 0). After we cleaned our data, our target variable is the same, but the name has been changed to “Depression”. It is our ethical duty to ensure this data was collected in a sensitive manner and this dataset adheres to these strict implications by considering privacy and informed consent.

2 Methods & Results

2.1 Exploratory Data Analysis

We performed exploratory data analysis on the original .csv file to clean it and identify ideas to explore in our statistical analysis. The first step was to ensure that there were no missing values. The next step was to rename column names to lower case with underscores instead of spaces. This gives consistency, an important factor when running analyses. An important part of the data cleaning was to understand the categorical variables. After plotting the distribution of these variables, there were some entries/values that overly complicated the data or were not relevant to our analysis. We simplified the data by dropping entries where the categorical variables were anachronistic to the majority of the data. For example, we chose to drop rows where the profession was not equal to student, and where students responded ”Other”. These variables did not add any clear meaning or were not in line with our focus on students. We also eliminated irrelevant columns such as city, degree, profession, and job satisfaction. After doing all of this, the indices were reset to ensure proper indexing after data was removed.

2.2 Research Goal: What Kinds of Students are at Risk for Depression?

2.2.1 Sleep & Depression

Question 1: Does Lack of Sleep Impact Rate of Depression?

Exploratory analysis revealed that the rate of depression is higher among students who get less than 5 hours of sleep per night vs. students who get more than 5 hours of sleep per night. The difference becomes more apparent when we consolidate the sleep duration categories into simply < 5 and > 5 hours.

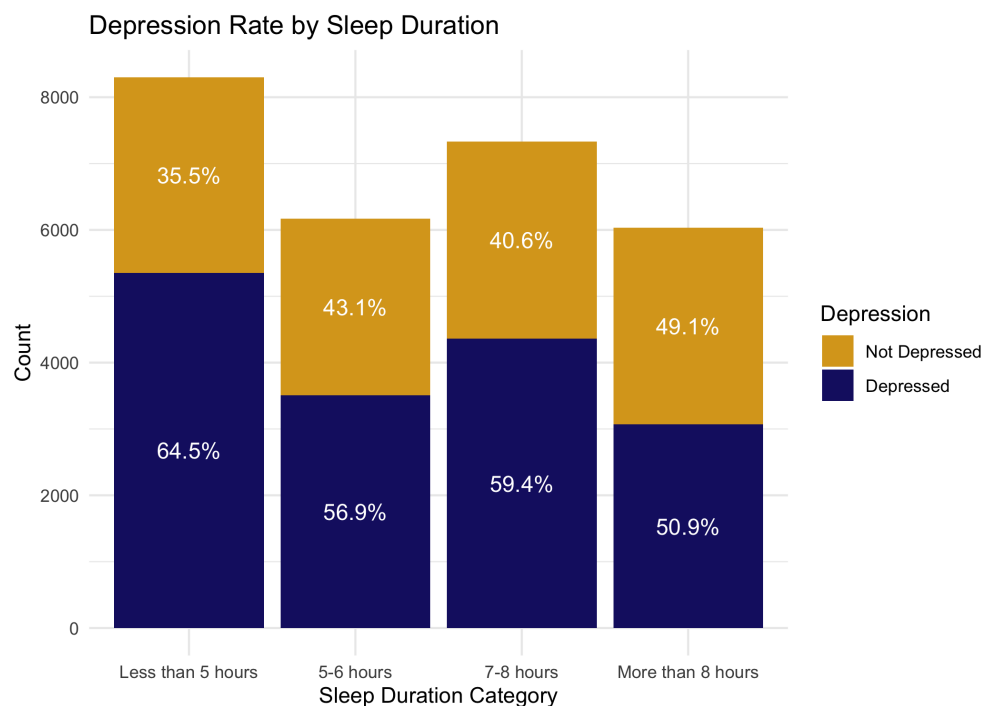


Figure 1: Depression by Daily Sleep Hours

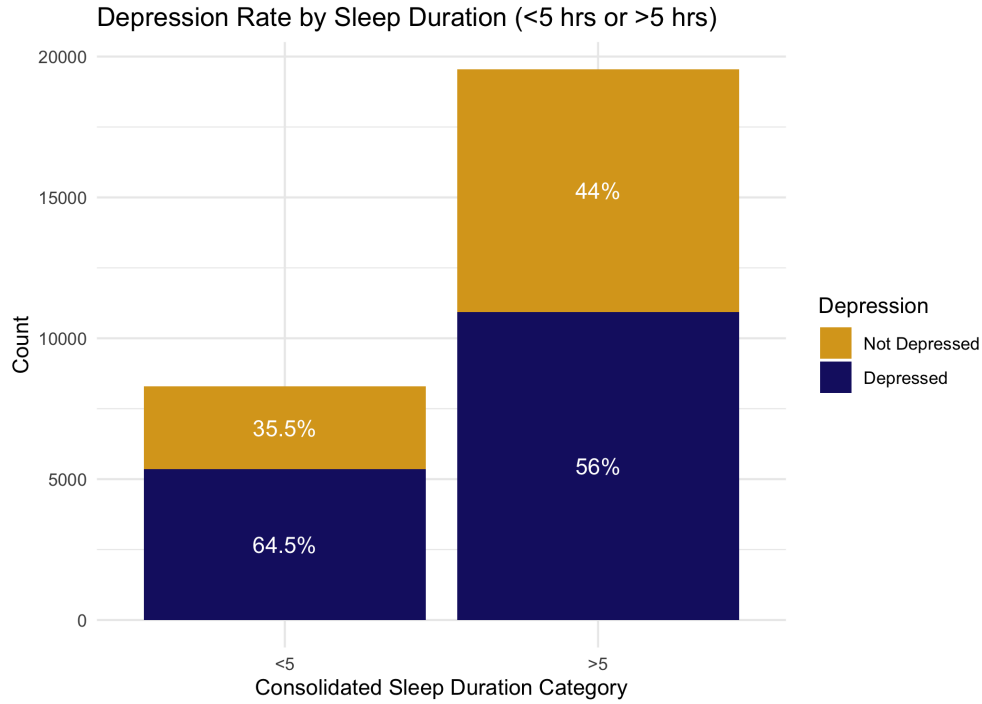


Figure 2: Depression by Daily Sleep Hours (Consolidated)

Approach & Analysis

Simple analysis revealed that:

$$P(\text{Depression} | < 5 \text{ hours of sleep}) \approx 0.6451$$

$$P(\text{Depression} | > 5 \text{ hours of sleep}) \approx 0.5598$$

Thus the two sample proportions are:

$$p_{<5} = 0.6451$$

$$p_{>5} = 0.5598$$

And the difference between them is:

$$p_{<5} - p_{>5} = 0.6451 - 0.5598 = 0.0853$$

Therefore in our sample dataset, students who sleep less than 5 hours per night have an approx. 8.53% higher rate of depression than students who sleep more than 5 hours per night. This lead to the development of null and alternate hypotheses to test whether students who sleep less than 5 hours per night have a statistically significantly higher level of depression than students who sleep more than 5 hours per night (significance level $\alpha = 0.01$).

$$H_0 : p_{<5} - p_{>5} = 0$$

$$H_A : p_{<5} - p_{>5} > 0 \quad (p_{<5} > p_{>5})$$

To test the null hypothesis it is appropriate to use a **2-proportion z-test**. We first performed a manual test, then used R's `prop.test()` for an automated approach.

Manual 2-Proportion Z-Test

Hours of Sleep/Night	n	Depressed	% Depressed
< 5	8,297	5,352	0.6451
> 5	19,540	10,938	0.5598
Total	27,837	16,290	0.5852

Table 1: Summary of Sleep/Depression Rates

From Table 1, we see the overall proportion $p = 0.5852$. This allows us to calculate the overall standard error using the equation:

$$\begin{aligned}
 SE &= \sqrt{p(1-p) \left(\frac{1}{n_{<5}} + \frac{1}{n_{>5}} \right)} \\
 &= \sqrt{0.5852 \times (1 - 0.5852) \left(\frac{1}{8927} + \frac{1}{19540} \right)} \\
 &= 0.00645
 \end{aligned}$$

We then calculate the z-statistic:

$$\begin{aligned}
z_{stat} &= \frac{p_{<5} - p_{>5}}{SE} \\
&= \frac{0.6451 - 0.5598}{0.00645} \\
&= 13.209
\end{aligned}$$

This yields an extremely low p-value which makes sense given the very large value of z_{stat} :

$$\begin{aligned}
p - value &= 1 - \phi(z_{stat}) \\
&= 1 - pnorm(z_{stat}) \\
&= 1 - pnorm(13.209) \\
&\approx 0
\end{aligned}$$

A p-value of approx. 0 is less than our significance level $\alpha = 0.01$, indicating that there is significant statistical evidence that $p_{<5}$ is greater than $p_{>5}$, or that students who sleep less than 5 hours per night have a higher incidence of depression than students who sleep more than 5 hours per night.

We confirm this using R's `prop.test()`:

```

2-sample test for equality of proportions with continuity correction

data: counts out of n
X-squared = 174.13, df = 1, p-value < 2.2e-16
alternative hypothesis: greater
99 percent confidence interval:
 0.07044064 1.00000000
sample estimates:
 prop 1    prop 2 
0.6450524 0.5597748

```

Figure 3: 2-Proportion Z-Test for Sleep Hours

This tests outputs a p-value of 4.637×10^{-40} , which is dramatically less than $\alpha = 0.01$. Ergo, we reject the null hypothesis that students who sleep less than 5 hours per night have the same rate of depression than students who sleep more than 5 hours per night. There is statistically significant evidence that $p_{<5} > p_{>5}$.

2.2.2 Suicidal Thoughts & Gender

Approach & Analysis: Question 2 and 3

For these questions, I decided to use Welch's two-sample t-test for means and a two-sample z-test for proportions. Both factors, depression and suicidal thoughts are measured with binary variables (Yes = 1, No = 0). Since the mean of the binary variables show a proportion we can conduct these tests, especially since our sample size is reasonably large. Our method also meets all assumptions for these tests: our observations are independent of each other, the Central Limit Theorem states that sample sizes larger than 30 are approximately normal, and Welch's t-test does not assume equal variance between our groups.

Question 2: Is there statistically significant evidence that there is a difference in suicidal thoughts between gender?

$$H_o : \mu_{male} = \mu_{female}$$

$$H_a : \mu_{male} \neq \mu_{female}$$

Welch's T-Test:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$\bar{x}_1, \bar{x}_2$ = sample means of female's and male's suicidal thoughts score

s_1, s_2 = sample variances of female's and male's suicidal thoughts score

$n_1, n_2 = 27, 837$

Two Sample Z-Test for proportions:

$$z = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\hat{p}(1 - \hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

$\hat{p}_1, \hat{p}_2$ = sample proportions of female's and male's suicidal thoughts score

$\hat{p}$ = pooled sample proportions of "Yes" across both male and females

n_1, n_2 = sample size

We also wanted to see if there was any difference in means between males and females for suicidal thoughts. For this, we used another two-sample t-test and two sample z-test. In the t-test, since our p-value of 0.8133 is greater than our alpha value of 0.05, we fail to reject the null hypothesis. We can say that there is not statistically significant evidence to say that there is a difference in means for suicidal thoughts between gender. The mean for suicidal thoughts for females is $\tilde{0}.6334$ and for males, $\tilde{0}.6320$.

```
Welch Two Sample t-test

data:  suicidal_thoughts_num by gender
t = 0.23613, df = 26441, p-value = 0.8133
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
 -0.01002784  0.01277486
sample estimates:
mean in group Female    mean in group Male
      0.6333766          0.6320031
```

Figure 4: Welch's Two Sample t-test: Suicidal Thoughts and Gender

For the two-sample proportion test, I reached the same conclusion, failure to reject the null hypothesis. The p-value of 0.8231 is much higher than our alpha value of 0.05. The proportion of suicidal thoughts for females is $\tilde{0}.4419$ and for males is $\tilde{0}.4433$. There is not enough evidence to show that there is a difference in mean or proportions of depression score between males and females. The confidence interval does have zero within it, which also shows there is insignificant statistical evidence for a difference of proportions.


```

2-sample test for equality of proportions with continuity correction

data:  table(df$suicidal_thoughts_num, df$gender)
X-squared = 0.049996, df = 1, p-value = 0.8231
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.01363786  0.01072167
sample estimates:
   prop 1    prop 2 
0.4418696 0.4433277

```

Figure 5: Two-sample z-test: Suicidal Thoughts and Gender

2.2.3 Gender & Depression

Question 3 : Is there statistically significant evidence that there is a difference in depression scores between genders?

$$H_o : \mu_{male} = \mu_{female}$$

$$H_a : \mu_{male} \neq \mu_{female}$$

Welch's T-Test:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$\bar{x}_1, \bar{x}_2$ = sample means of female's and male's depression score

s_1, s_2 = sample variances of female's and male's depression score

$n_1, n_2 = 27,837$

Two Sample Z-Test for proportions:

$$z = \frac{(\hat{p}_1 - \hat{p}_2)}{\sqrt{\hat{p}(1 - \hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

$\hat{p}_1, \hat{p}_2$ = sample proportions of female's and male's depression score

$\hat{p}$ = pooled sample proportions of "Yes" across both male and female

$$n_1, n_2 = 27,837$$

In testing the difference in means of depression score between female and male, we fail to reject the null hypothesis. Since our p-value of 0.7302 is greater than our alpha value of 0.05, we can say that there is not statistically significant evidence to say that there is a difference in means for depression between genders. The mean for depression score for females is $\tilde{0}.5840$ and for males, $\tilde{0}.5861$.

```
Welch Two Sample t-test

data:  depression by gender
t = -0.34482, df = 26424, p-value = 0.7302
alternative hypothesis: true difference in means between group Female and group Male is not equal to 0
95 percent confidence interval:
 -0.01370412  0.00960370
sample estimates:
mean in group Female    mean in group Male
      0.5840500          0.5861002
```

Figure 6: Welch's Two Sample t-test: Gender and Depression

I also conducted a two-sample proportion test, where I reached the same conclusion: a p-value not within the rejection region and a failure of the null hypothesis. The proportion of suicidal thoughts for females is $\tilde{0}.4440$ and for males is $\tilde{0}.4420$. There is not enough evidence to show that there is a difference in mean or proportions of depression score between males and females. The 95% confidence interval also proves this finding, since zero is within the confidence interval.

```
2-sample test for equality of proportions with continuity correction

data:  table(df$depression, df$gender)
X-squared = 0.11063, df = 1, p-value = 0.7394
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.009834373  0.014002110
sample estimates:
 prop 1    prop 2 
0.4440114 0.4419276
```

Figure 7: Two-sample z-test: Gender and Depression

2.2.4 Family History & Depression

Approach & Analysis

We decided to use a Chi-squared test of independence because we wanted to see if two categorical variables had association with one another. We chose family history because there have been many studies debating whether family history plays a role in mental illness.

Question 4 : Is there statistically significant evidence that there is a relation between mental illness in family history and depression?

H_o : Depression is independent of family history

H_a : Depression is dependent of family history

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

O_{ij} = number of students who experience one of these categories: family issues and depression, family issues and no depression, no family issues and depression, no family issues and no depression.

E_{ij} = hypothetical number of students who would fall into those categories if family history and depression were independent.

Pearson's Chi-squared test with Yates' continuity correction

```
data: tab
X-squared = 79.033, df = 1, p-value < 2.2e-16
```

Figure 8: Chi-squared: Family History and Depression

Since our p-value is less than alpha of 0.05, we can reject the null hypothesis. The p-value, 2.2e-16, falls within the rejection region with 0.05 as alpha. There is statistically significant evidence of an association between family issues and depression.

3 Conclusions

From our statistical analysis, we had many findings that were significant. The difference of proportions regarding sleep and depression showed that sleep and depression are related with those that sleep five hours per night having an approximately 8.53% higher risk of having depression. Then a two-proportion z-test was conducted, based upon those results comparing those that sleep under five hours to those that sleep more than five hours. We were able to reject the null hypothesis that there is no difference in depression rates for those that sleep over five hours compared to those sleeping more than five hours.

From our testing and confidence intervals, for Questions 2 and 3, we cannot say that there is difference in means and proportions for depression scoring between males and females. Likewise, there is not significant evidence to prove that there is a difference for means and proportions between suicidal thoughts and gender. Our findings suggest that gender has no relevance on the rate of depression or suicidal thoughts.

The Chi-squared test revealed there is an association between family history of mental illness and depression. This did not reveal anything that surprised us. From these tests we are able to make an educated assumption that sleep and family history are both risk factors when it comes to depression.

In regards to future research, we would like to see analysis involving factors of individuals' past experiences and further research into family history. It would be interesting to see if family history and depression is only related when family history refers to mental illness as a whole or if this would stand true for specific mental illnesses. In addition, since sleep is related to depression, it would be important to go further with this research and compare sleep quality with depression rather than just amount of sleep.